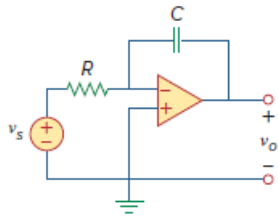


Problem

For the integrator shown in Fig. 10.112, obtain V_o/V_s . Find $v_o(t)$ when $v_s(t) = V_m \sin \omega t$ and $\omega = 1/RC$.

Figure 10.112:



Step-by-step solution

Step 1 of 3

Refer to Figure 10.112 in the textbook.

Draw the circuit diagram in frequency domain.

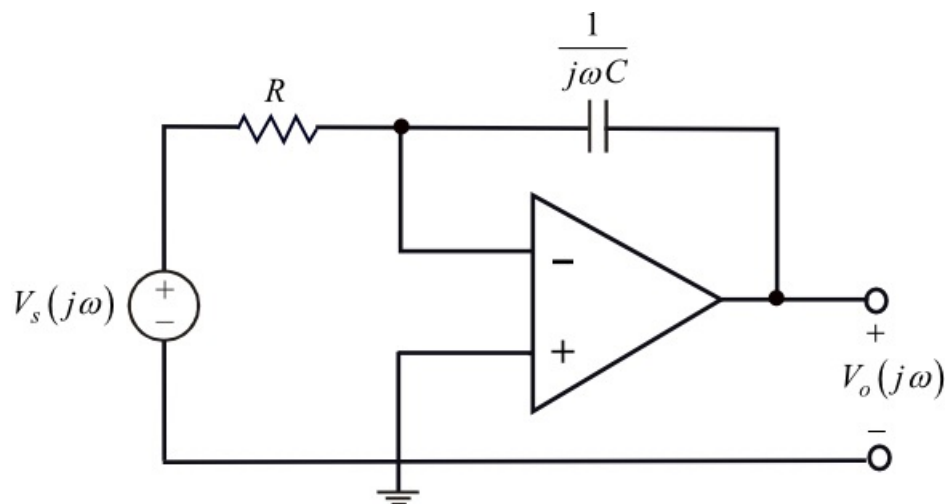


Figure 1

Step 2 of 3

From Figure 1, the feedback impedance is,

$$Z_f = \frac{1}{j\omega C}$$

The input impedance is,

$$Z_i = R$$

Since the circuit in Figure 1 is an inverting amplifier, the closed loop gain is,

$$\begin{aligned}\frac{V_o(j\omega)}{V_s(j\omega)} &= -\frac{Z_f}{Z_i} \\ &= -\frac{\frac{1}{j\omega C}}{R} \\ &= \frac{j}{\omega RC}\end{aligned}$$

Therefore, the closed loop gain, $\frac{V_o(j\omega)}{V_s(j\omega)}$ is $\boxed{\frac{j}{\omega RC}}$.

Step 3 of 3

Substitute $\frac{1}{RC}$ for ω in closed loop gain.

$$\begin{aligned}\frac{V_o(j\omega)}{V_s(j\omega)} &= \frac{j}{\frac{1}{RC} RC} \\ &= j1 \\ &= 1\angle 90^\circ\end{aligned}$$

Therefore,

$$v_o(t) = (1\angle 90^\circ) v_s(t) \dots\dots (1)$$

The input is,

$$v_s(t) = V_m \sin(\omega t)$$

From equation (1),

$$\begin{aligned}v_o(t) &= (1\angle 90^\circ) V_m \sin(\omega t) \\ &= V_m \sin(\omega t + 90^\circ) \text{ V}\end{aligned}$$

Therefore, the output voltage, $v_o(t)$ is $\boxed{V_m \sin(\omega t + 90^\circ) \text{ V}}$.